

Scoped evidence. Measured decisions.

Gemini development pilots and the path to an independent benchmark

Edition 0.4 • Observations through 29 September 2026 (Toronto)
Development report • Human interpretation and final review pending

Orbit evaluates atomic claims against exact passages and declared conditions. The milestone checks the extraction contract on six controlled questions and observes model-chosen Sanity reads. Schema, browser and citation-chain failures remain visible.

Observed layer	Measured result
Deterministic development suite	Baseline, N and P: 36/36 decisions; equality
Kaggle notebook pilot v2	Flash and Pro: 6/6 decisions, 8/8 exact passages, 53/53 checks
Antigravity pilot v1	Three runs: 6/6 decisions; one rejected scope record in each
Public agent observations	W01: consent refusal. W02: Flash/Pro read Context; provenance audit fails.

Interpretation: these measurements support development readiness and reveal a contract repair. Independent real-source validation and model repetitions will determine the product default. The current scores establish no theoretical superiority.

Proposed research-partner disclosure — author review pending

I used Codex to its fullest potential as a research partner—for code mapping, source comparison, evidence organization, consistency checks, editorial control, and deliverable preparation. I formulated the intent, defined the scope, interpreted the results, arbitrated the conclusions, and preserved every public decision. This collaboration expands my investigative capacity; judgment, responsibility, authorship, and final signature remain under my authority.

01 / A common evidence contract

What is classified

The unit is an atomic claim in a declared scope, rather than a whole document labeled true or false. A passage may support a claim, oppose another claim, or provide context. Scope carries subject, property, value and the applicable provider, product, execution mode, conditions or version.

Output	Meaning
T — support	Verified direct passages supporting the claim in scope
I — indeterminacy	Explicit missing conditions, unavailable evidence or unchecked passages
F — counterproof	Verified direct passages opposing the same claim in scope
ADMIT / REJECT / HOLD	Operational recommendation, separate from human judgment

T, I and F remain independent evidence sets. A conflicting claim can have both T and F. Their counts are neither probabilities nor three portions summing to 100%. Presence of an exact quotation and semantic relevance are separate questions.

Why pilot v2 differs from v1

The fifth v1 case asks about an unspecified retention condition. Each tested model returns the appropriate HOLD and an exact contextual quote, but provides a scope object missing property and value. The portable schema rejects that record. We preserved the original outputs.

Prompt v2 states the two valid forms explicitly: a complete atomic scope, or an omitted scope for purely contextual evidence. Both Kaggle model runs satisfy the clarified contract. This is a new prompt version; v1 and v2 scores are kept separate.

How leakage is controlled

Models receive the public twelve-document input and six questions. Gold labels are used by the scorer. The same extraction is fed to all engines. The final campaign will freeze independent cases, source snapshots, rules and code hashes before scoring.

02 / Results and their limits

Model / host	Prompt	Decision	Quote	Schema
Gemini 3.8 Flash / Kaggle	v2	6/6	8/8	0 errors
Gemini 3.1 Pro Preview / Kaggle	v2	6/6	8/8	0 errors
Gemini 3.8 Flash High / Antigravity UI	v1	6/6	8/8	1 error
Gemini 3.8 Flash High / Antigravity CLI	v1	6/6	8/8	1 error
Gemini 3.1 Pro High / Antigravity CLI	v1	6/6	8/8	1 error

In Kaggle, all three engine configurations evaluate six records correctly for each model. In v1, each engine evaluates five records correctly and rejects the remaining malformed scope. Denominators remain visible so that a rejected record cannot disappear from the reported coverage.

Assertions are checks, not sample size

The 53 Kaggle assertions cover six questions, quote checks, schema validity and engine outputs. They are correlated checks within the same small sample. The development suite also reuses known synthetic cases. These results cannot support a generalization claim or a scientific ranking.

Equal decisions have a specific explanation

The classic baseline and N currently share eligibility and recommendation logic. N represents independent support, counterproof and uncertainty; that representation alone does not create a different decision algorithm. P additionally applies discrete attribute rules. An independent corpus must isolate the cases where those rules change a conclusion correctly.

Resources and host differences

Kaggle notebook usage reports 1,147 input tokens for each model; Flash outputs 2,173 tokens and Pro 5,261. Their reported quota costs sum to US\$0.074435. The later task-page quota reads US\$0.18 used of US\$10 daily and US\$100 monthly; it includes a broader activity scope. These are existing free quotas, with no purchase or top-up.

Antigravity CLI uses the existing official account and its actual High model variants. Host defaults, caching and reasoning budgets differ from Kaggle. Durations and tokens remain descriptive metadata; they are not a normalized intelligence or speed ranking.

03 / What actually ran on Orbit

Native WebMCP transport

An isolated headed Chrome session discovers and invokes tools registered by the live page. The scripted transport check observes fifteen tools, READY capabilities, and CONSENT_REQUIRED when requesting an unshared private question. It does not manufacture dossier consent or human decisions.

Public Sanity Context reads

Actual public-page reads returned READY for privacy and deep-research paths, and for Sanity Knowledge Base and schema/Studio paths. The returned text comes from the existing Knowledge Base. This proves connectivity; the effect of its structure on a full answer still requires the paired Context-versus-text experiment.

Model-chosen calls through an external JSON planner

Flash chooses capabilities and a private-question request, then receives READY and CONSENT_REQUIRED from the live page. Its final answer overgeneralizes that refusal to Context: the public corpus has a separate permission boundary. Pro reads capabilities, then a page-generation guard refuses its next request. These observations do not repair native Antigravity MCP integration or establish a full research workflow. Earlier failures remain preserved.

W02: a response with an incomplete evidence chain

Flash makes four native calls, reads two entries separately and returns four exact quotes; only one claim includes an observed original URL. Its fourth quote also needs semantic review. Pro makes two calls and combines both entries despite the instruction: individual entry binding fails, both URLs are absent, and its limitations array is empty. Neither response passes the provenance audit.

Preserved attempt	Observed limitation
W02 startup interruptions	Three attempts stop before any model call; not model reasoning failures
W01 native CLI integration	No injected tools, prohibited fallback or filters; no completed native mission
Upstream Context citations	Named Web references have no original URL in the returned text

Orbit now reports observed citation metadata and unresolved URLs. It preserves the Context text and never infers a source address from a title. Reading an entry does not establish original-document verification or semantic support.

Release and startup verification

The preceding public release passed eight route checks and three byte-for-byte artifact checks, with a rollback backup. A headline startup race is fixed: an immediate initialize request now receives a response before further requests. The regression uses real native WebMCP and no model. This edition is prepared for a separately verified release.

04 / The independent campaign ahead

Layer	Acceptance evidence still required
Corpus	Ten packets × twelve documents; sixty questions; real labels human-reviewed
Engines and models	Common extraction; balanced repetitions; false contradictions and HOLD coverage
Sanity contribution	Same original corpus and question; structured Context versus text retrieval
Agent missions	Actual tool choice, bounded handoffs, provenance, revocation and stale revisions
Product review	Two human end-to-end paths, source audit and dossier export

The engine selection follows integrity first, fewer unjustified conclusions second, useful resolution third, then scope distinctions and cost. A simpler reference may win. The product retains N and P as explicit configurations while measurement determines their role.

Four real-source candidate packets now contain 48 retrieved documents and 24 questions: IA policies, Sanity documentation, robotic pollination, and swarms. None has a human-adjudicated gold answer yet. Metadata and links are public; full texts stay private pending license review.

Reproduction entry points

```
python tools/benchmark_suite_a.py --corpus .orbit/benchmark-corpus-development --output .orbit/benchmark-results/suite-a-replay.json

python tools/summarize_benchmark.py

python tools/test_webmcp_transport.py

python tools/prepare_kaggle_pilot.py --model google/gemini-3.8-flash --model google/gemini-3.1-pro-preview
```

Run canonical scripts from the Orbit repository root. New model calls require a quota precheck and a fresh experiment identifier. A saved trace is a replay, not a new independent run.

Inspect the evidence

[Live protocol, aggregates and private Kaggle owner links](#)

Frozen pilot fingerprints

```
Shared TypeScript bundle

886f7cf3fbfcb445f4c1af7ebf1d3f8c
b581964fe20d3bce706e8b4f4b880bd9

Prompt v2

785070d61245c64226ad070dff0196b3
473edde8644fc333c501342d807e9c5f
```

Canonical safe outputs, CSV, source hashes, failed attempts and harness snapshots are indexed under the existing .benchmark dossier. Public artifacts omit credentials and internal model reasoning. Publication and challenge submission remain separate future actions.